

results_for_ben

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Exits predicts wellbeing / loneliness

```
lmer(wmws ~ exits + gender + (1 | id), data = new_df_latent_vars) %>%  
summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: wmws ~ exits + gender + (1 | id)  
## Data: new_df_latent_vars  
##  
## REML criterion at convergence: 2177.9  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.2371 -0.5139  0.0550  0.5829  3.0185   
##  
## Random effects:  
## Groups   Name                Variance Std.Dev.  
## id      (Intercept)  0.2254     0.4748  
## Residual                    0.2792     0.5284  
## Number of obs: 1090, groups: id, 411  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)      
## (Intercept)  -0.04451    0.03538  403.48682  -1.258    0.209      
## exits         0.17249    0.01504 1053.51163  11.470   <2e-16 ***  
## gendermale    0.14099    0.06037  407.07594   2.336    0.020 *    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) exits  
## exits         0.010  
## gendermale    -0.586 -0.018
```

```
lmer(loneliness ~ exits + gender + (1 | id), data = new_df_latent_vars) %>%  
summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]
```

```
## lmerModLmerTest]
## Formula: loneliness ~ exits + gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 1636.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.2041 -0.5851 -0.0712  0.5385  3.2353
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.1665  0.4080
## Residual 0.1627  0.4033
## Number of obs: 1074, groups: id, 411
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -0.02743    0.02947   391.94468  -0.931    0.353
## exits       -0.10809    0.01208  1057.72673  -8.945   <2e-16 ***
## gendermale   0.09006    0.05011   394.89112   1.797    0.073 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) exits
## exits         0.017
## gendermale    -0.588 -0.027
```

Exits is moderated by time

```
lmer(wmws ~ exits * time + gender + (1 | id), data = new_df_latent_vars) %>%
summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: wmws ~ exits * time + gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 2158.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5372 -0.4915  0.0431  0.5691  2.8146
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.2326  0.4823
## Residual 0.2643  0.5141
## Number of obs: 1090, groups: id, 411
##
```

```
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.08767   0.04166  703.32242    2.104  0.0357 *
## exits         0.16434   0.02221 1020.30673    7.401 2.82e-13 ***
## times2       -0.22050   0.03889  725.77149   -5.669 2.07e-08 ***
## times3       -0.19368   0.03918  722.97093   -4.943 9.56e-07 ***
## gendermale    0.13862   0.06051  406.68834    2.291  0.0225 *
## exits:times2  -0.02919   0.02782  809.14742   -1.049  0.2943
## exits:times3   0.01517   0.02814  818.20868    0.539  0.5901
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) exits  times2 times3 gndrml exts:2
## exits      -0.100
## times2     -0.451  0.111
## times3     -0.452  0.108  0.496
## gendermale -0.503 -0.002 -0.001  0.013
## exits:tims2 0.071 -0.647 -0.015 -0.063 -0.006
## exits:tims3 0.080 -0.657 -0.065 -0.043 -0.016  0.555
```

```
lmer(loneliness ~ exits * time + gender + (1 | id), data = new_df_latent_vars) %>%
  summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: loneliness ~ exits * time + gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 1646.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.2851 -0.5741 -0.0987  0.5273  3.1662
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.1659  0.4073
## Residual          0.1610  0.4013
## Number of obs: 1074, groups: id, 411
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.010761  0.033924  657.529187    0.317  0.751177
## exits        -0.116021  0.017773  990.975367   -6.528 1.06e-10 ***
## times2       -0.015641  0.030598  697.988052   -0.511 0.609379
## times3       -0.102551  0.030895  693.619125   -3.319 0.000949 ***
## gendermale    0.086894  0.049980  395.083468    1.739 0.082891 .
## exits:times2   0.016793  0.022061  774.660957    0.761 0.446757
## exits:times3   0.000977  0.022391  781.774732    0.044 0.965209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##          (Intr) exits  times2 times3 gndrml exts:2
## exits      -0.081
## times2     -0.428  0.100
## times3     -0.430  0.094  0.490
## gendermale -0.513 -0.009 -0.005  0.014
## exits:tims2 0.061 -0.642 -0.004 -0.055 -0.009
## exits:tims3 0.065 -0.649 -0.051 -0.032 -0.013  0.557
```

Exits is moderated by sex

```
lmer(wmws ~ exits * gender + (1 | id), data = new_df_latent_vars) %>%
  summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: wmws ~ exits * gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 2182.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2235 -0.5096  0.0586   0.5849   3.0417
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.2259  0.4753
## Residual 0.2792  0.5284
## Number of obs: 1090, groups: id, 411
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.04434    0.03541 403.27489  -1.252  0.2112
## exits          0.17966    0.01859 1052.23291   9.667 <2e-16 ***
## gendermale     0.14152    0.06042 406.81544   2.342  0.0196 *
## exits:gendermale -0.02088    0.03165 1053.25072  -0.660  0.5095
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) exits  gndrml
## exits          0.012
## gendermale    -0.586 -0.007
## exts:gndrml   -0.007 -0.587 -0.013
```

```
lmer(loneliness ~ exits * gender + (1 | id), data = new_df_latent_vars) %>%
  summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
```

```
## Formula: loneliness ~ exits * gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 1642.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.1957 -0.5826 -0.0745  0.5322  3.2377
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.1668  0.4085
## Residual          0.1627  0.4034
## Number of obs: 1074, groups: id, 411
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.02724   0.02950  391.86189  -0.924  0.3563
## exits         -0.10363   0.01510 1054.01323  -6.864 1.14e-11 ***
## gendermale      0.09044   0.05015  394.70316   1.803  0.0721 .
## exits:gendermale -0.01233   0.02520 1058.95409  -0.489  0.6246
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) exits gndrml
## exits         0.021
## gendermale   -0.588 -0.013
## exts:gndrml -0.013 -0.599 -0.016
```

Does exits5 and exits8 predict wellbeing when controlling for exits

```
lmer(wmws ~ exits5 + exits8 + exits + gender + (1 | id), data = new_df_latent_vars) %>%
summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: wmws ~ exits5 + exits8 + exits + gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 1425.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.59318 -0.55356  0.06075  0.58401  2.04361
##
## Random effects:
## Groups Name Variance Std.Dev.
## id      (Intercept) 0.2460  0.4960
## Residual          0.2878  0.5365
## Number of obs: 668, groups: id, 361
```

```

##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -0.37630    0.10566 645.55109  -3.562 0.000396 ***
## exits5       0.01290    0.01548 662.46891   0.833 0.405122
## exits8       0.04122    0.01822 662.87944   2.263 0.023980 *
## exits        0.14607    0.02338 629.64957   6.248 7.67e-10 ***
## gendermale   0.17090    0.07053 352.92273   2.423 0.015900 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) exits5 exits8 exits
## exits5      -0.374
## exits8      -0.749 -0.198
## exits        0.478 -0.174 -0.434
## gendermale  -0.214 -0.074  0.026  0.002

lmer(loneliness ~ exits5 + exits8 + exits + gender + (1 | id), data = new_df_latent_vars) %>%
  summary()

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: loneliness ~ exits5 + exits8 + exits + gender + (1 | id)
## Data: new_df_latent_vars
##
## REML criterion at convergence: 1023.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.40049 -0.54522 -0.08671  0.50246  2.45854
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## id       (Intercept)  0.1694    0.4116
## Residual                0.1445    0.3802
## Number of obs: 655, groups: id, 360
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  0.189159   0.082256 647.238548   2.300 0.021786 *
## exits5      -0.003269   0.011639 638.355265  -0.281 0.778935
## exits8      -0.048371   0.013872 642.507401  -3.487 0.000522 ***
## exits       -0.086742   0.018020 637.823920  -4.814 1.85e-06 ***
## gendermale   0.089994   0.055504 355.941922   1.621 0.105821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) exits5 exits8 exits
## exits5      -0.377
## exits8      -0.757 -0.174
## exits        0.470 -0.180 -0.429
## gendermale  -0.228 -0.076  0.036 -0.008

```

Look at exits5 and exits8 descriptively

```
new_df_latent_vars %>%  
  select(time, exits5, exits8) %>%  
  pivot_longer(cols = c(exits5, exits8), names_to = "var", values_to = "val") %>%  
  mutate(val = mean(val, na.rm=TRUE), .by = c("time", "var")) %>%  
  ggplot(aes(x=time, y=val)) +  
  geom_point() +  
  geom_line() +  
  theme_classic() +  
  facet_wrap(~var, scale="free")
```

```
## Warning: Removed 834 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 834 rows containing missing values or values outside the scale range  
## ('geom_line()').
```

